

How Much of the Disruption Pile-Up Is a Routing Artifact?

A deterministic routing-ladder study on the PSC simulator (no RL)

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Summary

The $\sim 3,800$ -unit backlog that accumulates at the disrupted distributor was previously treated as unavoidable (supply through the disrupted chain is capped at ~ 20 /period). We tested that assumption by changing *only the order-routing rules* — no buffering, no anticipation, no learning — in a ladder of four configurations, scored at the **system level** (all six agents) on held-out seeds. **Result: roughly two-thirds of the during+post disruption cost is a routing artifact, not physics.** Flipping the equal-split health center to trust-based splitting removes 26%; adding a sharper split rule and a stale-order write-off removes 67%; adding an onset-triggered reroute removes **91%**. Under urgent demand, routing fixes alone prevent 71–87% of lost patients. During-disruption fill rate reaches 1.000: the healthy chain could serve all demand all along — the binding constraint was a routing rule, not supply. This means our earlier value-decomposition numbers were computed against a hobbled routing layer and must be recomputed, and the earlier “detection captures ≈ 0 ” claim narrows to *detect-then-buffer*: *detect-then-reroute* is the single most valuable deployable response measured to date.

1 Question and motivation

During the long disruption (periods 110–157), the manufacturer feeding $DS_{\text{disrupted}}$ drops to 5% capacity; deliveries through that chain are capped at ~ 20 /period against ~ 110 /period of demand. Prior analysis concluded the resulting pile-up was unavoidable: no ordering policy can pull more product through a capped chain.

That argument is correct *given the routing configuration* — but the configuration contains two suspect rules, both verified in code. First, HC_{equal} splits its orders **equally** between the two distributors by a fixed rule (`decision_maker.py:158`), so it keeps sending half its demand to the dead chain for all 48 periods. Second, orders the dead chain cannot fill sit in the HC’s *on-order* position indefinitely (no timeout), and the ordering formula subtracts on-order before splitting (`decision_maker.py:153`) — so stranded orders also **suppress re-ordering from the healthy chain**.

Why this gates the project framing: our earlier value decomposition attributed $-468k$ to “defending at all” (a standing buffer) and $-654k$ to “timing” (unreachable by detection). If a large fraction of the disruption cost is recoverable by fixing a routing rule, those numbers are conditional on a frozen

routing layer, and the deployable-lever story becomes three-fold: buffering, routing flexibility, and detection.

2 Experimental setup

Table 1: Setup. All simulations are rule-based (base-stock ordering at every echelon); the routing rule is the only thing that changes between runs.

Component	Value
Topology	2 MN $\rightarrow$ 2 DS $\rightarrow$ 2 HC; each HC dual-sourced from both DSs
Chains	MN1 $\rightarrow$ DS _{disrupted} ; MN2 $\rightarrow$ DS _{healthy} (no MN-level rerouting exists)
Demand (primary, urgent0)	non-urgent $\mathcal{N}(120, 5)$ per HC per period; urgent = 0
Demand (secondary, urgent20)	non-urgent $\mathcal{N}(120, 5)$; urgent = 20 (lost if unmet)
Costs	holding 1/unit, backlog 10/unit, revenue 10/unit
Disruption	MN1 at 5% capacity, periods 110–157 (48 of 300 periods)
Phases	pre 60–109, during 110–157, post 158–300 (warm-up excluded)
HC ordering	up-to – on-order – inventory + backlog, capped at $\max(2d, 120)$
Trust	EMA of on-time delivery rate, $\delta = 0.1$
Seeds	pre-registered: tuning = [1–10], reporting = [11–30], paired
Scoring	system-level cost (all six agents), holding vs. backlog decomposed

Table 2: The routing ladder. Each rung adds one mechanism.

Rung	Change
(a)	Baseline, exactly as configured in the repo: HC _{trust} splits by trust, HC _{equal} splits equally. Reproduces the prior base-stock reference bit-for-bit (gate 1).
(b)	HC _{equal} flipped to the same trust-based splitting HC _{trust} uses. The core test.
(c)	(b) + sharper redirection (trust ⁴ split weights, trust update $\delta' = 0.3$) + stale-order write-off: on-order entries older than $5 \times$ lead time are ignored by the ordering computation (accounting-only; the ledger is untouched, late deliveries still arrive and are costed honestly). Parameters tuned on tuning seeds only; the ranking was stable across the whole grid.
(d)	(b) + oracle onset detection: from period 110 the HC split steps to a 90/10 share toward DS _{healthy} , reverting to trust dynamics after 157. This is <i>detect-then-reroute</i> — distinct from the <i>detect-then-buffer</i> response that earlier analysis showed captures ≈ 0 .

Six verification gates passed before any reported number: (1) rung (a) under constant demand reproduces the prior base-stock reference with *zero* difference over 300 periods; (2) per-seed determinism; (3) rung (b) measurably shifts HC_{equal}’s split; (4) the write-off measurably reduces counted on-order; (5) per-period demand conservation at both HCs; (6) cross-seed demand variation is nonzero. Gate 6 exists because we found `NormalDistPatientModel.__init__` hard-resets the global RNG (`patient_model.py:28`), which had silently made every “seed” the same demand path — a finding that also affects any historical multi-seed result that used the normal demand model. Our runner re-seeds after construction.

3 Results

Table 3: Main results, urgent0 config, held-out reporting seeds 11–30 (mean over 20 seeds; paired SEs are $\leq 0.3\%$ of the mean everywhere, so they are omitted from the table). TTR = time-to-recovery (periods after disruption end until $DS_{\text{disrupted}}$ backlog stays within 110% of its pre-disruption mean); AUB = area under the $DS_{\text{disrupted}}$ backlog curve, during+post (threshold-free).

	(a) baseline	(b) trust-split	(c) +sharp+w/o	(d) +onset
During+post system cost	3,700,784	2,730,635	1,209,659	332,166
vs. baseline	—	− 26.2%	− 67.3%	− 91.0%
during / post	2.58M / 1.13M	2.10M / 0.63M	0.95M / 0.26M	0.20M / 0.14M
Pre system cost (sanity)	17,044	17,040	17,045	17,040
Peak backlog $DS_{\text{disrupted}}$	4,067	2,877	1,004	168
Peak backlog DS_{healthy}	139	199	572	528
During fill rate (aggregate)	0.808	0.897	1.000	1.000
HC _{trust} / HC _{equal}	0.90 / 0.72	0.90 / 0.90	1.00 / 1.00	1.00 / 1.00
TTR $DS_{\text{disrupted}}$ (periods)	18	13	7	38.9 [†]
AUB $DS_{\text{disrupted}}$, during+post	131,253	93,526	33,797	4,126
<i>urgent20 config (same seeds):</i>				
During+post system cost	3,908,652	2,865,974	1,339,087	447,451
Lost patients (episode)	1,768	1,594	510	236
vs. baseline	—	−10%	− 71%	− 87%

[†]The 110%-threshold misbehaves for (d): its pre-disruption backlog is ≈ 0 , so the threshold is tiny and the rung is penalized for clearing a small residual slowly. The threshold-free AUB row is the honest comparison.

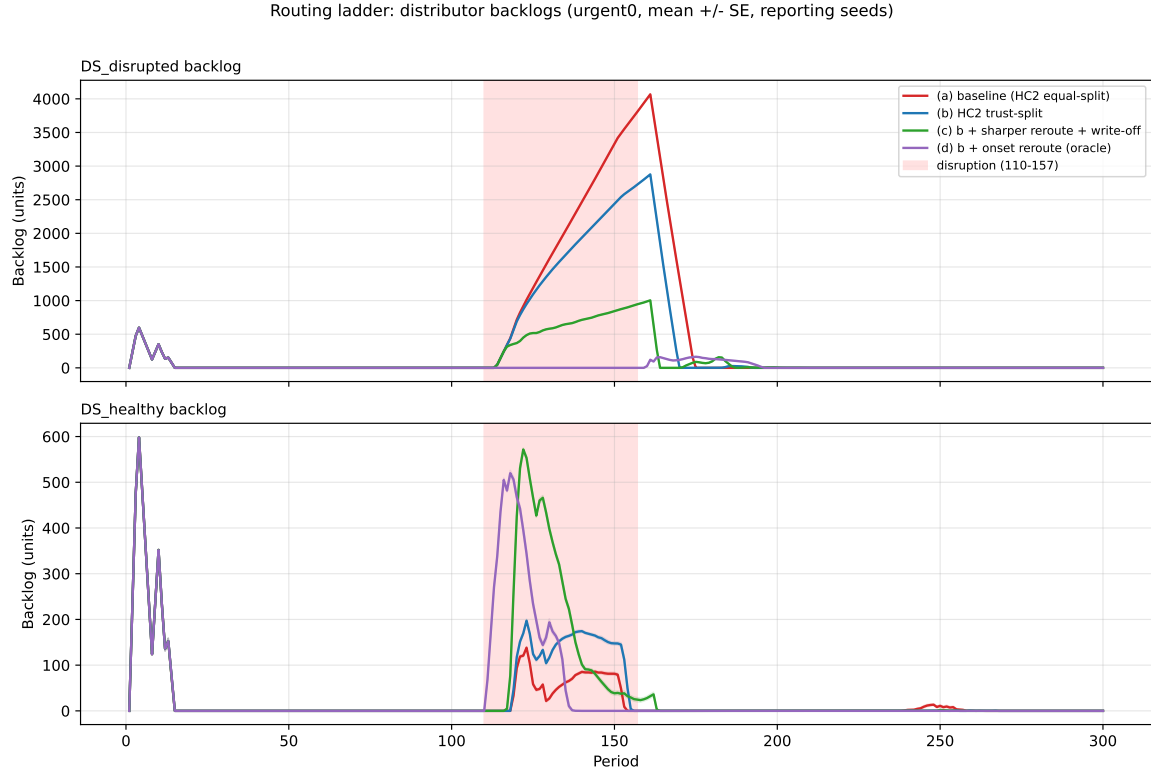


Figure 1: Distributor backlogs by rung (urgent0, mean $\pm$ SE over reporting seeds). The baseline pile-up at $DS_{\text{disrupted}}$ (top, red) shrinks rung by rung; DS_{healthy} (bottom; note the $6\times$ smaller scale) absorbs the redirected demand with only a brief transient — no mirror-image pile-up forms.

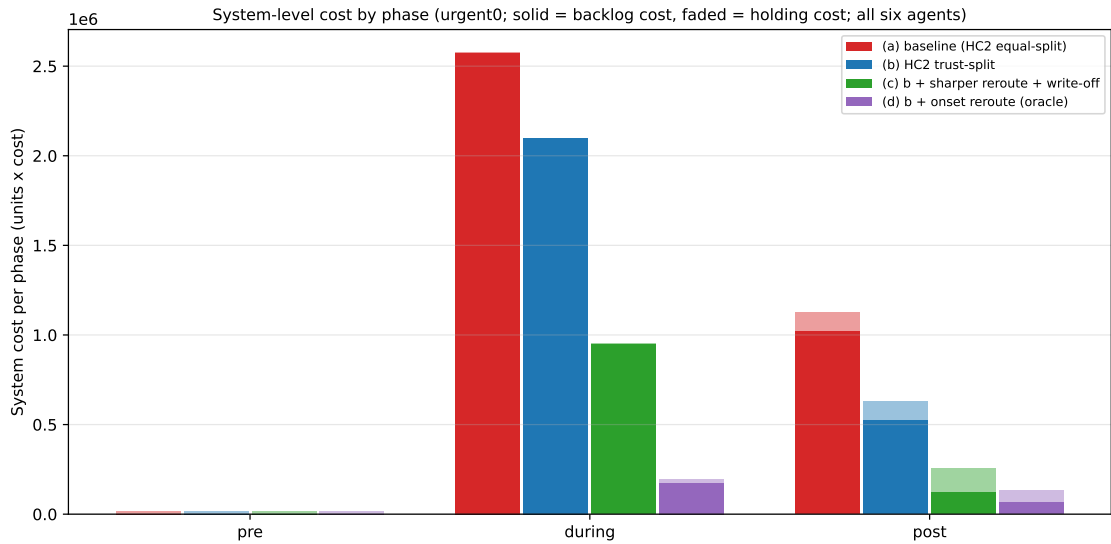


Figure 2: System-level cost by phase (solid = backlog cost, faded = holding cost, all six agents). Pre-phase costs are identical across rungs (no rung does anything before the disruption); the entire effect is during+post. Rung (c)'s slightly higher post holding is the write-off's late-delivery glut, honestly costed and dwarfed by its backlog savings.



Figure 3: The mechanism, directly: HC_{equal} 's share of orders sent to the dead chain. Baseline (red) stays at 0.5 by rule. Plain trust-split (blue) redirects slowly and bottoms out at ~ 0.23 — the trust-EMA floor predicted in the hypothesis doc. The sharper rule (green) reaches ~ 0.12 within ~ 10 periods; the onset reroute (purple) steps there immediately. All revert after recovery.

4 Interpretation and caveats

The healthy chain could serve everything all along. Under rungs (c)/(d) the during-disruption fill rate is 1.000: MN2 (capacity 400/period, normally loaded ~ 120) absorbs all redirected demand. The system was never supply-capped — only the disrupted *chain* was. “The pile-up is unavoidable” was true only chain-locally.

Each mechanism contributes measurably. Equal-split lock-in costs 26%; trust’s slow, incomplete redirection plus stranded-order suppression another 41 points; detection timing the last 24. The fairness effect is also fixed: baseline harm concentrates on HC_{equal} (fill 0.72 vs. 0.90), and every fixed rung equalizes the HCs.

What this means for our earlier numbers. The value decomposition (-468k buffer / -654k timing) was measured with the routing layer frozen in its hobbled state; it must be recomputed on a routing-fixed baseline. The “detection captures ≈ 0 ” result was real but response-specific: buffering a supply-capped chain post-onset is futile, while rerouting demand post-onset is worth $\sim 0.88\text{M}$ (the $c \rightarrow d$ gap) plus half of rung (c)’s remaining lost patients. The *anticipation* impossibility result stands untouched: nothing is observable before period 110, and rung (d) uses an oracle onset signal, so (d) is an upper bound on real detect-then-reroute. Since (c) reaches the redirected state within ~ 10 periods with no detector at all, a real detector with a few periods of latency lands between (c) and (d).

Caveats. The Doroudi et al. (2020) non-monotonicity prior (trust-sensitive buyers can be *disadvantaged* in some regimes) did not bind here — rung (b) improved every metric — but we tested one disruption regime; severity/duration robustness is queued. Results are one topology ($2 \times 2 \times 2$, healthy chain has ample headroom); in a regime where the healthy chain *cannot* absorb total demand, buffering should re-emerge as the binding lever. Rung (c)’s parameters were tuned on separate seeds, and the tuning ranking was stable, but the rule forms (trust^p, age write-off) were chosen by us, not optimized over rule space.

5 Next steps and future directions: simple heuristic policies

The following is assessment, not measured result, except where numbers are cited.

Immediate next experiments (all deterministic, unblocked):

1. **Recompute the value decomposition on the rung-(c) baseline** — rerun the buffer sweep and clairvoyant ceiling with routing fixed. This replaces the $-468\text{k}/-654\text{k}$ numbers and decides how much room is left for *any* anticipatory policy. Highest priority.
2. **Real detector in rung (d)** — swap the oracle for the delivery-shortfall detector from the earlier detection study; measure what detection latency costs against the (c)–(d) gap.
3. **Buffer $\times$ routing interaction** — $\{\text{no buffer, always-defend}\} \times \{\text{rung a, rung c}\}$: is buffering still worth anything once routing is fixed? (Prediction: little, at this severity — fill rate is already 1.000 under (c).)
4. **Robustness** — short/moderate disruptions, 50–80% capacity cuts, and a severity where DS_{healthy} *cannot* absorb everything.

Which simple heuristics look promising (for the heuristic-frontier paper):

- **Trust-sharpening (trust^p) and faster trust updates** — two-line change, no detection, captured -41 points here. The single best value-per-complexity lever found so far.
- **Stale-order write-off ($\text{age} > k \times \text{lead time}$)** — deployable accounting rule; in real supply chains this is just order-expiry. Part of the same -41 points.
- **Detect-then-reroute** — the one detection-based policy with measured value. Pairs a trivial onset detector (delivery shortfall) with a demand-side response.
- **Post-disruption catch-up throttle** — cap the order rate while clearing backlog, to suppress the recovery glut (visible in rung (c)’s post holding). Untested; cheap to add to the ladder.
- **Serve-captive allocation at the disrupted DS** — with routing fixed, the disrupted DS’s trickle should arguably go to whichever HC remains stuck with it. Tests dossier Mechanism 3 without any RL.
- **Permanent buffer** — now demoted to the lever of last resort for regimes where the healthy chain lacks headroom; its standing holding cost only pays off where routing cannot.

For the eventual RL comparison, this study sharpens the bounding-paper structure: simple rules now occupy -67% (no detection) to -91% (oracle detection) of the achievable range, so any learned policy must be evaluated against the rung-(c) baseline, not against the broken-routing baseline — and the residual available to learning is the gap between rung (c) and the (to-be-recomputed) clairvoyant ceiling.